Custom CNNs vs. Transfer Learning

Experiment Report — Food-101 Subset (10 Classes)

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Dataset Food-101 — 10 selected classes	**Transfer Model** MobileNetV2 (ImageNet weights)
Scratch Optimizer RMSprop (lr = 0.001)	**TL Optimizer** Adam (lr = 0.001)

1. Final Accuracy Comparison

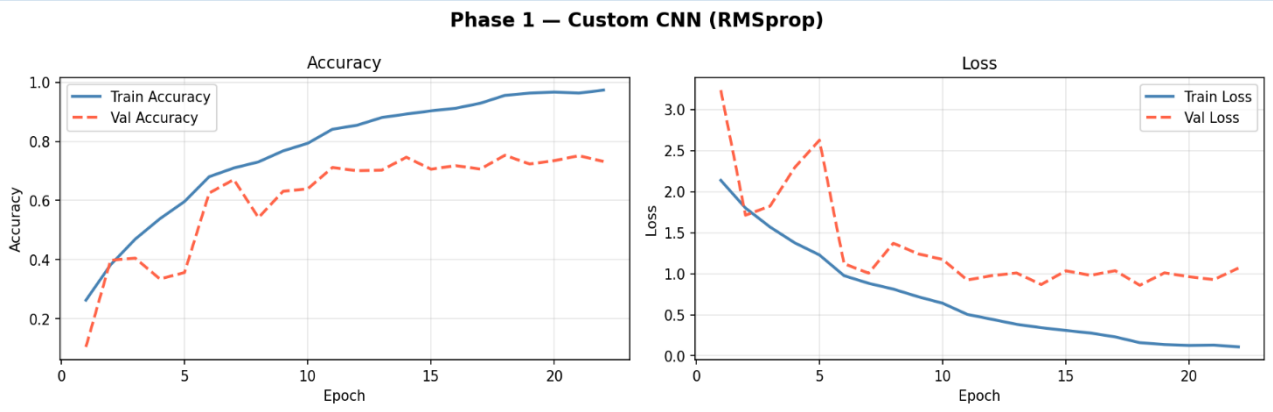
The table below summarises the key performance metrics recorded at the end of training for both models. Replace the placeholder values with your actual results from the notebook output.

Metric	Custom CNN (Scratch)	MobileNetV2 (Transfer)
Validation Accuracy	75.32%	88.48%
Total Epochs Run	22	11
Epochs to ≥50% Val Acc	6	1
Final Validation Loss	0.8602	0.3632
Optimizer	RMSprop (lr=0.001)	Adam (lr=0.001)
Trainable Parameters	457,738 p	12,810 p

2. Training Visualizations

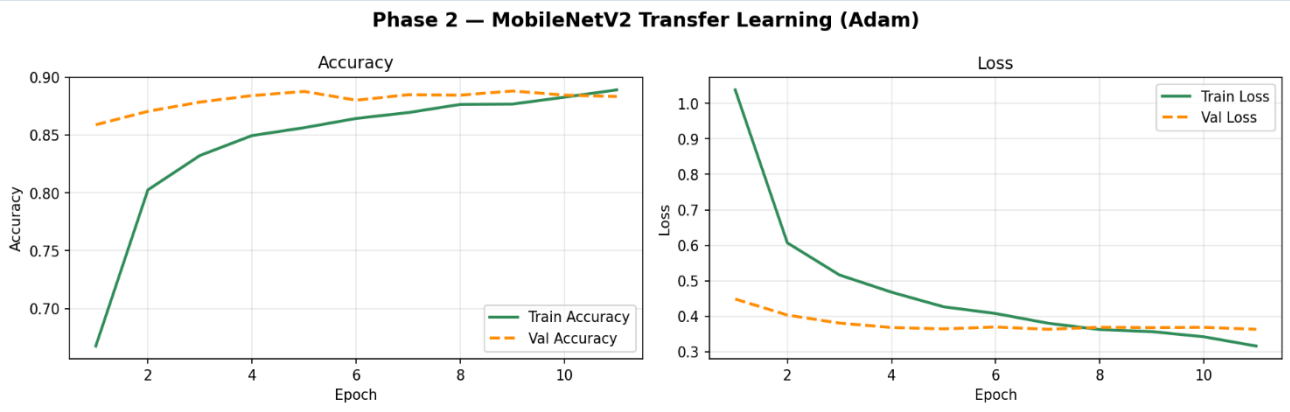
2.1 Phase 1 — Custom CNN (RMSprop)

Training and validation accuracy/loss curves for the custom CNN trained from scratch on the Food-101 10-class subset.



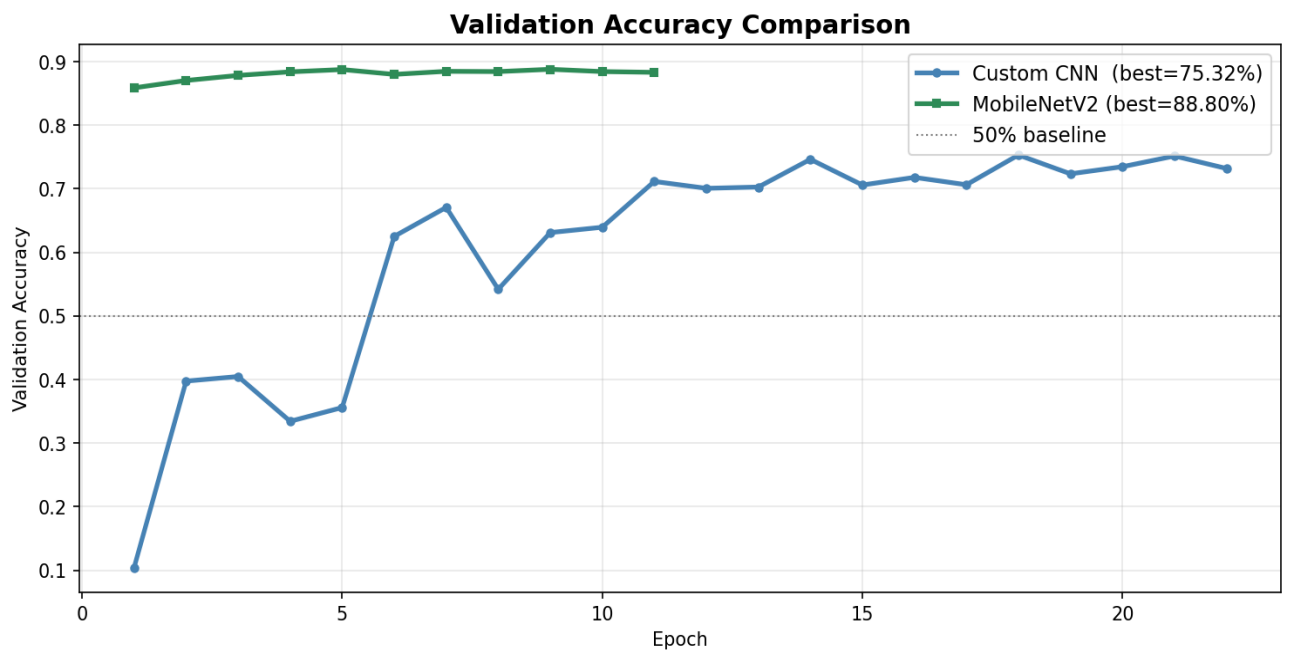
2.2 Phase 2 — MobileNetV2 Transfer Learning (Adam)

Training and validation accuracy/loss curves for the MobileNetV2 model with frozen base and trainable classification head.



2.3 Validation Accuracy — Head-to-Head

Direct comparison of validation accuracy across epochs for both models.



3. Conclusion

Why did the pre-trained model perform better?

Food-101 presents a highly complex visual classification challenge. Even when limited to 10 classes, each category exhibits enormous intra-class variation in lighting, plating style, angle, and ingredient arrangement. The following factors explain the performance gap:

- Feature richness of ImageNet pre-training: MobileNetV2 was trained on over 1.2 million images across 1,000 diverse categories. Its convolutional layers have already learned robust low-level (edges, textures) and mid-level (shapes, patterns) features that transfer directly to food recognition tasks.

- Data efficiency: The custom CNN must learn all feature representations from ~750–1,000 training images per class. This is insufficient for a deep network to generalise without overfitting, even with Dropout and data augmentation.

- Parameter efficiency: With the MobileNetV2 base frozen, only the lightweight classification head is trained. This dramatically reduces the risk of overfitting and means that convergence requires far fewer epochs.